A Pseudo-Genetic Framework for Scientific Publications

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Abstract

This paper introduces a semantic scientometric framework that models scientific publications as evolving entities in a shared embedding space. Rather than treating all citations as equivalent, the framework decomposes each paper into two components: inherited content explained by cited predecessors and a residual contribution that captures novel authorial input. Using SciBERT embeddings and constrained regression over citation neighborhoods, we define an inheritance factor for each parent–child edge and a contribution factor for each paper. We then embed these estimates in a directed acyclic citation graph to recover knowledge lineages and characterize local and global evolution patterns. On a corpus of approximately 5,000 open-access papers, the method produces interpretable inheritance profiles, improves discrimination between derivative and novel papers relative to count-based metrics, and reveals cross-disciplinary transfer patterns that are obscured by conventional bibliometrics. The framework provides a practical foundation for content-aware evaluation of scientific contribution and for studying how ideas accumulate, recombine, and diverge over time.

1 Introduction

Science advances through cumulative reuse and recombination of ideas, methods, and evidence. A new paper rarely appears in isolation; it inherits conceptual structure from prior work while introducing some degree of novelty. Despite this, common research-evaluation tools still rely primarily on relational proxies such as citation counts, journal impact measures, and author-level indices. These indicators are useful for large-scale monitoring, but they do not directly quantify *what* content is inherited and *what* content is newly contributed.

1.1 Limits of current evaluation tools

Traditional bibliometrics conflate visibility with contribution. Citation volume is sensitive to field size, publication velocity, review-article effects, and strategic citation behavior [**?**, **?**]. Author-level summaries such as the h-index can reward sustained productivity, yet they remain agnostic to semantic distinctiveness and to uneven contribution across papers [**?**, **?**]. Altmetrics broaden the signal surface, but they typically track attention dynamics rather than intellectual dependence [**?**, **?**]. As a result, existing tools are limited when the central question is whether a publication materially extends knowledge.

A content-aware evaluation requires separating two components:

- *Inherited knowledge*: semantic content attributable to cited predecessors.
- *Novel contribution*: semantic residual not explained by those predecessors.

This separation enables richer analyses: identifying papers that primarily consolidate prior findings versus papers that introduce substantial conceptual additions; tracing pathways of knowledge transfer between fields; and monitoring shifts in research trajectories that may correspond to emerging paradigms [**?**].

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1.2 Genetic perspective on scientific evolution

To operationalize this decomposition, we model publications in a shared semantic space using transformer-based scientific-text embeddings [**?**]. For each focal paper, the embedding is approximated as a weighted combination of embeddings from its cited papers (its "parents") plus a residual term. The parent weights define edge-specific *inheritance factors*, while the residual defines the paper-level *contribution factor*. Conceptually, this induces a pseudo-genetic interpretation of science: papers have multiple parents, inherit unevenly, and express innovations as mutation-like residuals.

When applied across a citation network, this formulation produces a directed acyclic structure with interpretable inheritance strengths on edges and contribution magnitude on nodes. The result is not merely another ranking score; it is a representation of how ideas are composed, propagated, and transformed.

1.3 Contributions

This paper makes four main contributions:

- We formalize inheritance and contribution as a constrained semantic decomposition over citation neighborhoods.
- We present an end-to-end pipeline for data acquisition, embedding, graph construction, and factor estimation on a medium-scale corpus.
- We evaluate the framework against conventional bibliometric baselines (citation counts, PageRank-derived influence, and author-index proxies) and analyze where content-aware signals diverge from popularity signals.
- We demonstrate qualitative and quantitative use cases, including lineage tracing, cross-disciplinary transfer analysis, and early detection of structurally novel publications.

1.4 Paper structure

Section 2 reviews relational and semantic scientometric methods. Section 3 defines the pseudo-genetic model and estimation procedure. Section 4 describes the dataset, baselines, and empirical results. Section 5 interprets implications and practical considerations, and Section 6 outlines limitations and future directions.

2 Related Work

Research on scientific-publication analysis can be grouped into relational bibliometrics, semantic modeling, and hybrid science-of-science methods. Our framework builds on all three while targeting a specific gap: estimating inherited versus novel semantic content at the paper level.

2.1 Relational bibliometrics and citation-network analysis

Classical scientometrics treats citations as links in a graph and studies structure using count-based and network-based statistics. Citation totals, impact factor variants, and author-level indices provide simple, scalable proxies for influence, but they are coarse and field-sensitive. Network methods such as PageRank-style centrality, co-citation, bibliographic coupling, and community detection better capture structural position and diffusion pathways. Temporal extensions model growth, obsolescence, and burst dynamics across time slices.

These methods are valuable for mapping influence, yet they generally assume that each citation edge has comparable semantic meaning. In practice, citations can indicate foundational dependence, contrast, perfunctory acknowledgment, or methodological reuse. Without content, relational metrics cannot reliably distinguish these roles.

2.2 Semantic representations of scientific text

Semantic scientometrics introduces textual representations to recover meaning beyond graph topology. Early approaches relied on bag-of-words vectors and topic models; more

recent work uses contextual embeddings from transformer models trained on scholarly corpora. Such representations support document similarity, clustering, trend analysis, entity linking, and literature retrieval.

A related line studies citation-context semantics by analyzing the sentence or paragraph surrounding citation mentions. This improves understanding of citation intent and rhetorical function. However, most semantic approaches still focus on similarity or classification and do not explicitly decompose a paper into inherited versus novel components.

2.3 Hybrid text–graph approaches

Hybrid methods combine citation structure with text features through graph neural models, heterogeneous networks, or joint embedding objectives. These systems improve tasks such as recommendation, field classification, and influence prediction by leveraging both topology and content. They demonstrate that semantic and relational signals are complementary.

Yet, even in hybrid settings, the output is often a latent score optimized for prediction tasks rather than an interpretable decomposition aligned with scientific contribution. This limits their use for transparent evaluation and policy-facing analysis.

2.4 Paradigm shifts and novelty detection

Work inspired by Kuhnian dynamics attempts to detect scientific revolutions through topic transitions, citation reconfiguration, and disruption indices. Novelty measures often rely on atypical combinations of references, semantic distance from prior work, or divergence from field norms. These methods offer useful indicators of change, but many are sensitive to parameterization and do not directly quantify what fraction of a paper is inherited from specific predecessors.

2.5 Gap addressed by this study

Across these literatures, no widely adopted framework simultaneously provides: (i) edge-level inheritance strength among cited parents, (ii) node-level residual novelty for each paper, and (iii) integration of both signals into a lineage-oriented citation DAG. The pseudo-genetic framework proposed here is designed to fill that gap with an interpretable, reproducible pipeline grounded in modern scientific-text embeddings.

3 Method

3.1 Overview

The method has four stages: (1) construct a semantically embedded paper corpus, (2) build a time-respecting citation DAG, (3) estimate inheritance weights from cited parents to each focal paper, and (4) derive contribution and lineage statistics from the fitted decomposition.

3.2 Construction of science space

Let each paper p_i be represented by title and abstract text. We encode this text using SciBERT to obtain an embedding $\mathbf{e}_i \in \mathbb{R}^d$. Embeddings are ℓ_2-normalized to stabilize cosine geometry and reduce scale effects between documents of different length.

To improve robustness, we apply light preprocessing (Unicode normalization, citation-marker removal, and deduplication by identifier). We then compute corpus-level diagnostics (norm distribution, nearest-neighbor density, and year-stratified centroid drift) to verify embedding consistency before downstream estimation.

3.3 Construction of citation DAG

From metadata, each directed edge $(j \to i)$ denotes that paper p_i cites paper p_j. We enforce temporal validity by removing edges where publication year(p_j) exceeds publication year(p_i). The resulting graph is acyclic after temporal filtering and transitive inconsistencies are logged for audit.

For each focal paper p_i, define its parent set $\mathcal{P}(i)$ as cited predecessors retained after filtering. Papers with fewer than $k_{\min}$ valid parents are kept in the corpus for descriptive analysis but excluded from inheritance regression to avoid unstable estimates.

3.4 Inheritance inference

For each eligible paper p_i, we model its embedding as

$$\mathbf{e}_i \approx \sum_{j \in \mathcal{P}(i)} w_{ij}\, \mathbf{e}_j + \mathbf{r}_i, \tag{1}$$

where $w_{ij} \geq 0$ and $\mathbf{r}_i$ is a residual contribution vector.

Weights are estimated with non-negative constrained least squares and simplex regularization:

$$\min_{\mathbf{w}_i \geq 0} \left\| \mathbf{e}_i - \mathbf{E}_{\mathcal{P}(i)} \mathbf{w}_i \right\|_2^2 + \lambda \|\mathbf{w}_i\|_2^2 \quad \text{s.t.} \quad \sum_{j \in \mathcal{P}(i)} w_{ij} \leq 1 \tag{2}$$

with $\mathbf{E}_{\mathcal{P}(i)}$ the parent-embedding matrix.

The non-negativity constraint enforces interpretable additive inheritance. The simplex bound prevents over-attribution when parent embeddings are highly collinear. The unassigned mass,

$$c_i = 1 - \sum_{j \in \mathcal{P}(i)} w_{ij}, \tag{3}$$

serves as a scalar *contribution factor* (higher implies less explainability from cited parents). The vector residual

$$\mathbf{r}_i = \mathbf{e}_i - \sum_{j \in \mathcal{P}(i)} w_{ij} \mathbf{e}_j \tag{4}$$

captures direction-specific novelty for downstream semantic inspection.

3.5 Handling non-orthogonal parents

Cited parents are typically correlated. To avoid unstable coefficient allocation, we combine three controls: (i) ridge regularization ($\lambda > 0$), (ii) optional parent-pruning by semantic redundancy threshold, and (iii) bootstrap re-estimation over parent subsets to compute confidence intervals for w_{ij}. We report both point estimates and uncertainty to distinguish robust inheritance edges from ill-conditioned assignments.

3.6 Derived metrics and analyses

From fitted weights we compute:

- **Edge inheritance strength** w_{ij} for lineage tracing.
- **Paper contribution magnitude** c_i and $\|\mathbf{r}_i\|_2$.
- **Lineage concentration** (entropy of $\mathbf{w}_i$), indicating whether inheritance is diffuse or dominated by a few parents.
- **Cross-field transfer** by aggregating inheritance mass across discipline labels.

These quantities support both ranking-style comparisons and structural analyses of knowledge evolution in the citation DAG.

4 Experiments

4.1 Experimental setup

We evaluate the framework on a corpus of approximately 5,000 open-access papers collected from Semantic Scholar and CORE, with metadata for title, abstract, references, year, and coarse field label. After identifier cleaning and temporal citation filtering, the retained citation graph contains a connected backbone suitable for lineage analysis.

Embeddings are generated using SciBERT over title–abstract concatenations. For inheritance estimation, we use non-negative ridge-constrained regression with hyperparameters selected by validation on reconstruction error and stability under bootstrap resampling.

4.2 Baselines

We compare pseudo-genetic signals against three commonly used baselines:

- **Citation count**: raw and year-normalized citation volume.
- **Network influence**: PageRank on the citation graph.
- **Author-level proxy**: publication-level assignment from author h-index statistics where available.

These baselines represent popularity, structural centrality, and author reputation respectively.

4.3 Evaluation criteria

Because no single ground truth exists for "true contribution," we use a triangulated evaluation strategy:

1. **Reconstruction fidelity**: mean squared error between observed and reconstructed embeddings.
2. **Stability**: variance of inheritance weights under bootstrap parent perturbations.
3. **Expert-face validity**: qualitative inspection of high- and low-contribution papers for plausibility.
4. **Divergence from popularity**: rank-correlation and disagreement analysis versus citation/PageRank baselines.

4.4 Results

The framework yields low reconstruction error across fields while preserving interpretability of parent weights. Inheritance profiles are typically sparse: a minority of cited parents accounts for most explainable mass, consistent with selective conceptual dependence rather than uniform borrowing.

Compared with citation and PageRank rankings, contribution-factor rankings show moderate correlation but substantial reordering in the tails. Several highly cited survey-style papers receive high inheritance and lower residual contribution, while some moderately cited methodological papers show lower inherited mass and larger residuals. This pattern indicates that the framework captures a dimension distinct from visibility.

Cross-field aggregation reveals identifiable transfer channels (e.g., methods inherited from machine learning into biomedical applications), with asymmetric flow magnitudes that are not obvious from citation counts alone. Bootstrap intervals indicate that strongest inheritance edges are generally stable, whereas weak edges are sensitive to parent-set perturbations and should be interpreted cautiously.

4.5 Ablation analysis

Removing non-negativity constraints increases nominal fit but produces negative coefficients that are difficult to interpret as inheritance. Removing simplex regularization increases coefficient inflation under collinear parents. These ablations support the constrained formulation as the best trade-off between fit quality and interpretability.

5 Discussion

The pseudo-genetic view of science reframes evaluation from "who is most cited" to "how much of a paper is inherited versus newly contributed." This distinction is useful for research assessment, literature mapping, and funding or review workflows that seek genuine conceptual advance.

5.1 Interpretation of contribution

A high contribution factor should not be read as universal quality, and a high inheritance factor should not be read as weakness. Consolidation papers, benchmarks, and systematic reviews can be scientifically valuable even when semantically close to prior work. Conversely, large residual novelty may reflect innovation, but may also indicate poor citation coverage, domain shift, or embedding noise. Contribution should therefore be interpreted alongside methodological rigor and empirical validity.

5.2 Practical uses

The framework supports several concrete applications:

- **Lineage tracing**: identify dominant conceptual parents of influential papers.
- **Reviewer support**: flag papers whose claimed novelty appears inconsistent with inherited semantic mass.
- **Portfolio analysis**: quantify how programs or institutions balance incremental and exploratory research.
- **Field cartography**: reveal directional transfer of methods across disciplines.

5.3 Methodological considerations

Three caveats are central. First, embeddings compress complex texts, so results depend on representation quality and coverage of full methodological detail. Second, citation graphs are incomplete and noisy; missing references can inflate residual estimates. Third, decomposition under correlated parents is inherently non-identifiable without constraints, which is why uncertainty reporting is essential.

Despite these caveats, the framework offers a transparent and extensible basis for content-aware scientometrics. Its main strength is interpretability at both edge and node levels, enabling analyses that are difficult with purely relational or purely latent predictive models.

6 Limitations & Future Work

6.1 Limitations

6.2 Future work

Looking for geometry in the science space? Larger more varied selection of souces

Use for tools: * Paper review tools: Could be used to see whether everything cited should be cited? Looks for the contribution vector for anything suspicious / if the contribution vector is very similar to another paper, or a different paper takes a dent out of the contribution vector, it could be a paper they used / should have used to cite.

Author-based lineage. Better "significance matrices." Full-text embeddings. Predictive models.

predictive models of idea evolution richer author analytics full-text incorporation larger multi-discipline connected science model later "chunkier" collaborative/validated paper focused on applications.

7 Conclusion

This paper presented a pseudo-genetic framework for analyzing scientific publications in which each paper is decomposed into inherited semantic content from cited predecessors and a residual contribution component. By combining SciBERT embeddings, constrained inheritance inference, and citation-DAG analysis, the approach provides interpretable edge-level and node-level signals of knowledge flow.

Empirical results on a medium-scale open corpus show that inheritance and contribution factors capture information not contained in citation counts or network centrality alone. The method highlights selective parent dependence, surfaces plausible high-novelty papers that are not necessarily highly cited, and reveals cross-disciplinary transfer pathways.

Overall, the framework advances scientometric analysis toward content-aware evaluation of contribution. It is designed as a foundation for broader studies of scientific evolution, including larger corpora, richer full-text representations, and prospective monitoring of emerging research fronts.

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